

~~L4~~

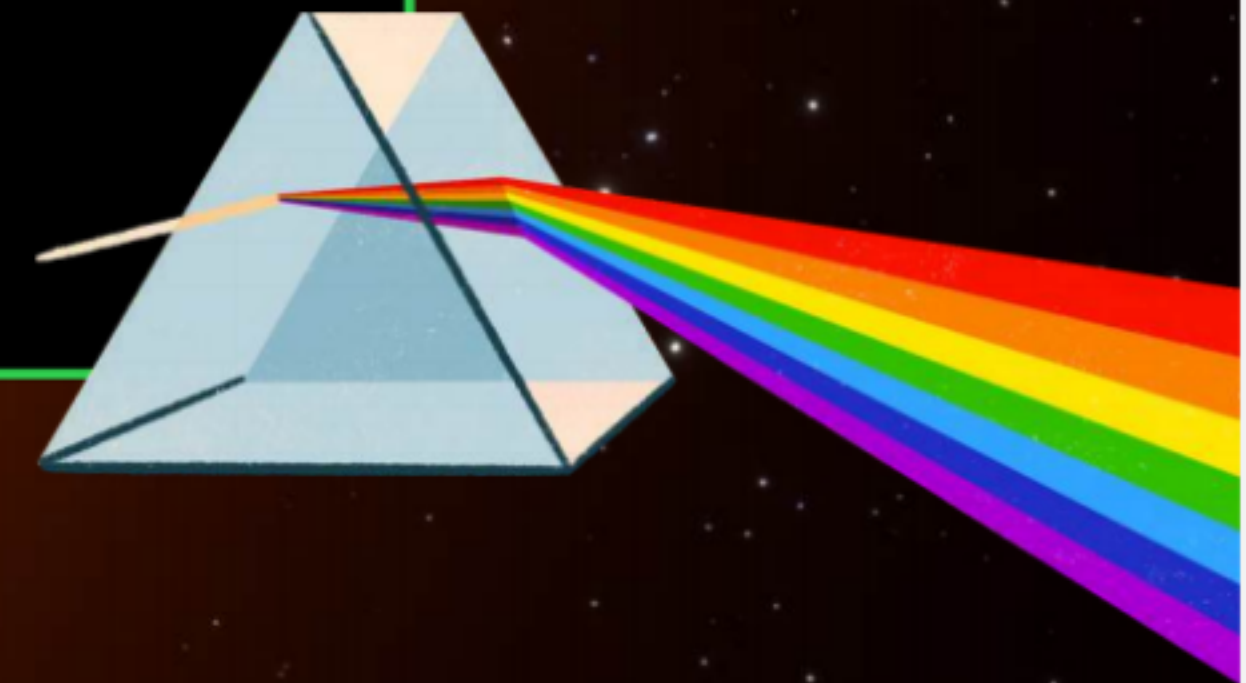
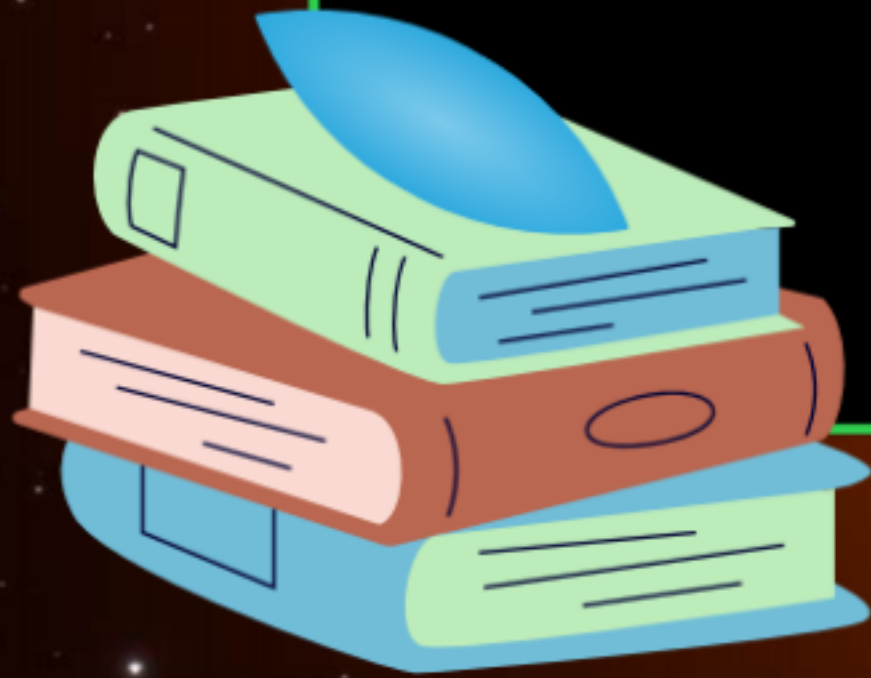
L5

आरंभ
BATCH

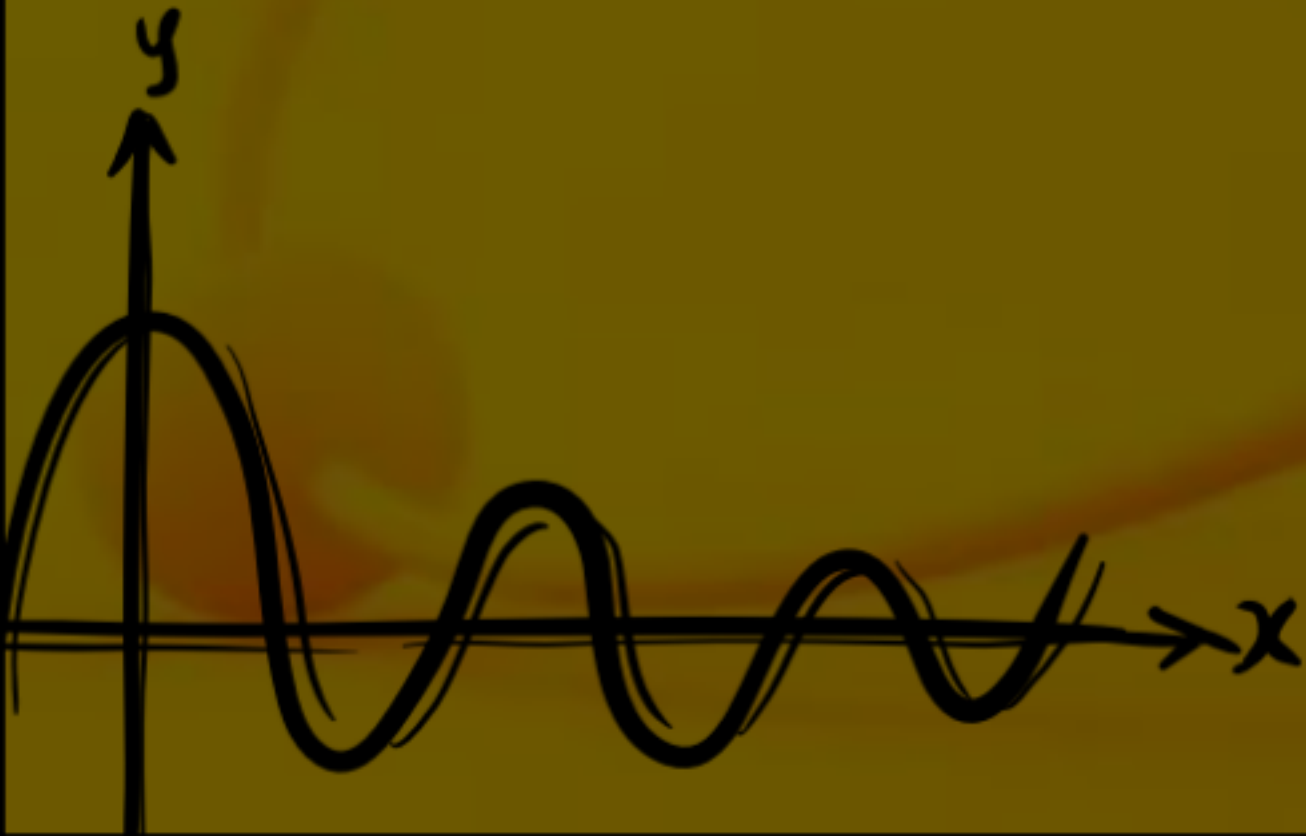
CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD

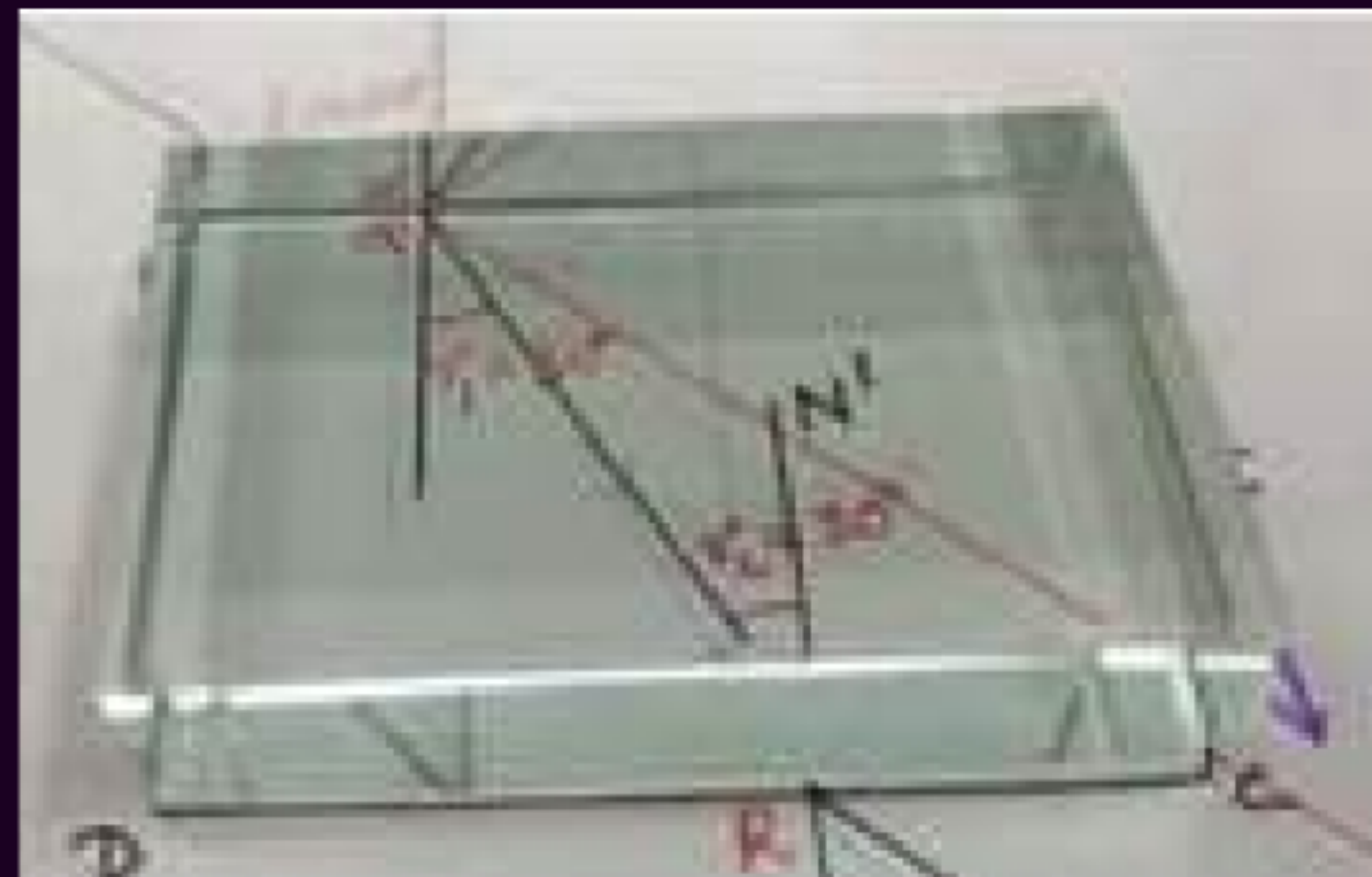


Refraction





Pencil Appears Bent in Water

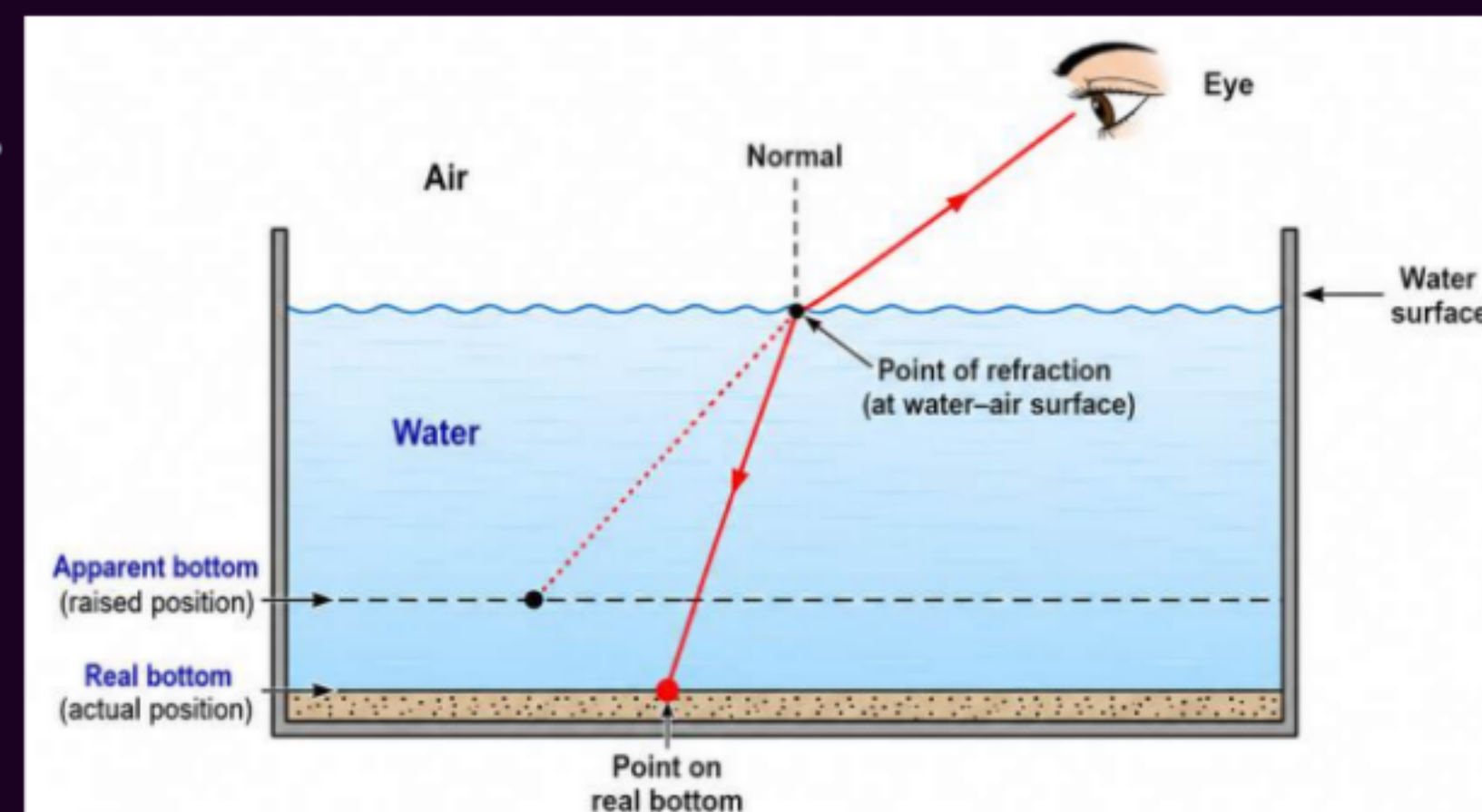


Letters Appear Raised Through Glass Slab



Lemon Appears Bigger in Water

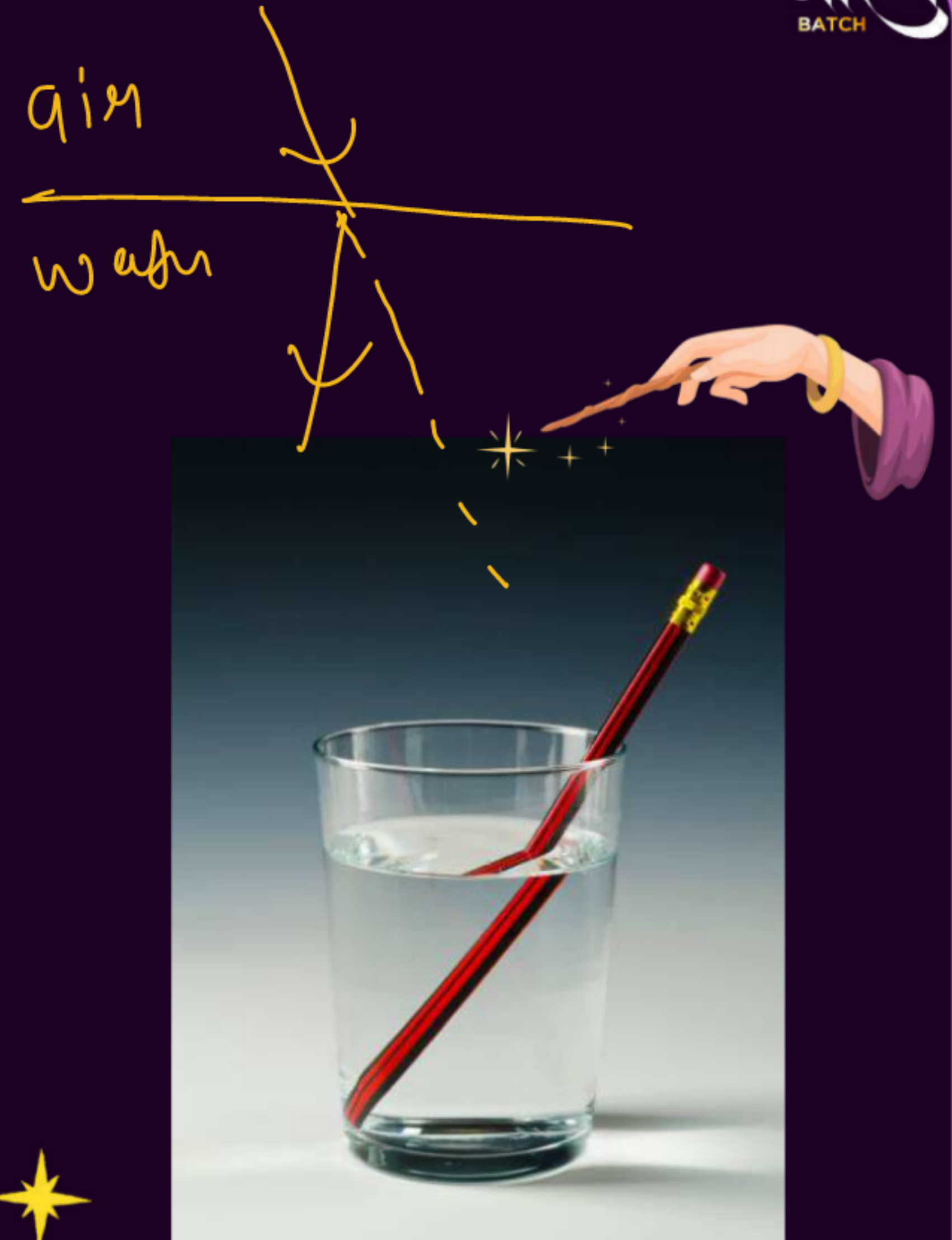
Bottom of Tank Appears Raised



Refraction

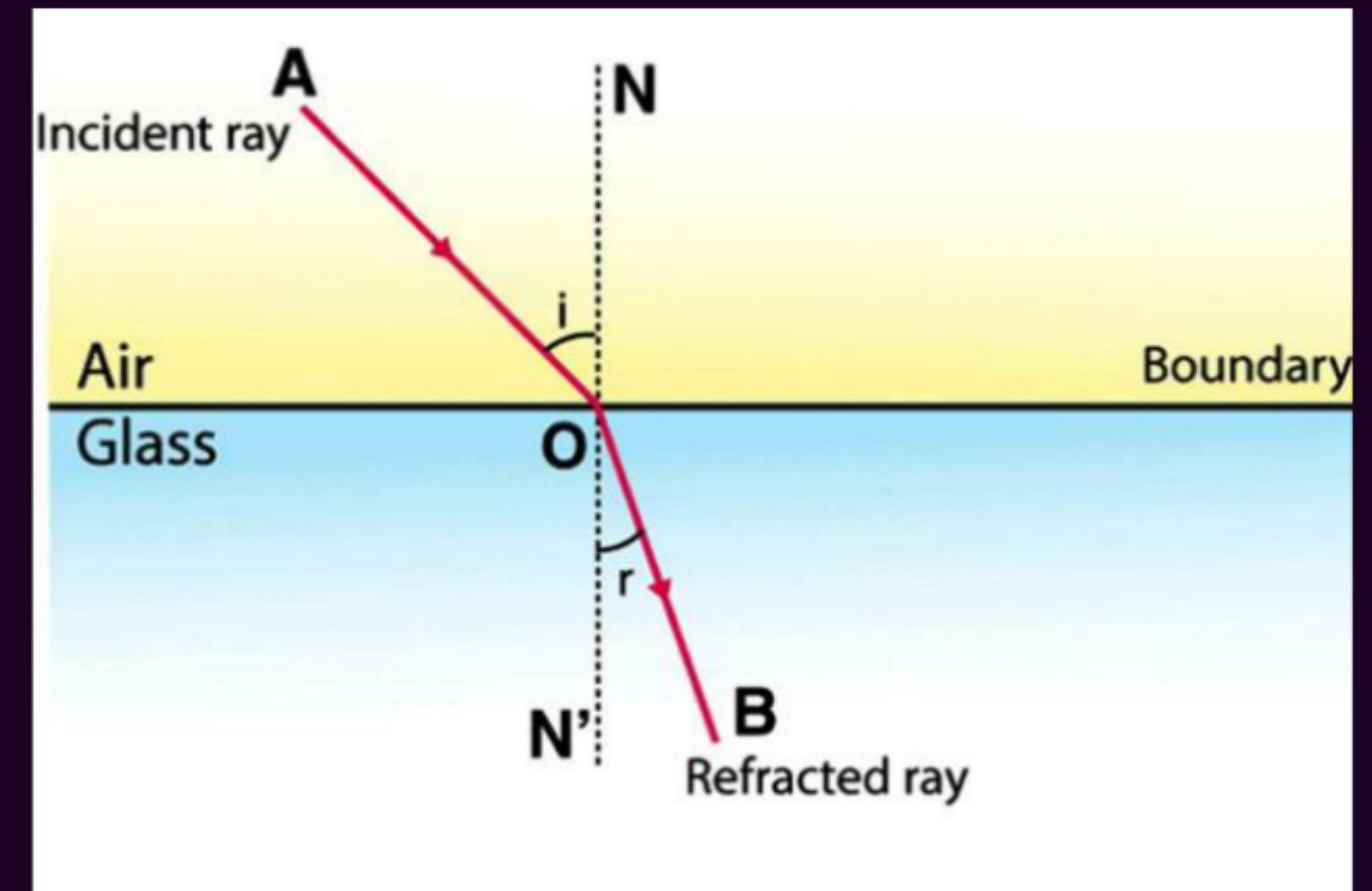
Refraction is the bending of light when it travels from one medium to another due to change in its speed.

This change in direction happens because light travels at different speeds in different media.



Causes of Refraction

- When the light goes from air into water, it bends towards normal because there is a **reduction in its speed**.
- When the light goes from water to air, it bends away from normal because there is an **increase in the speed of light**.



Medium



Rarer

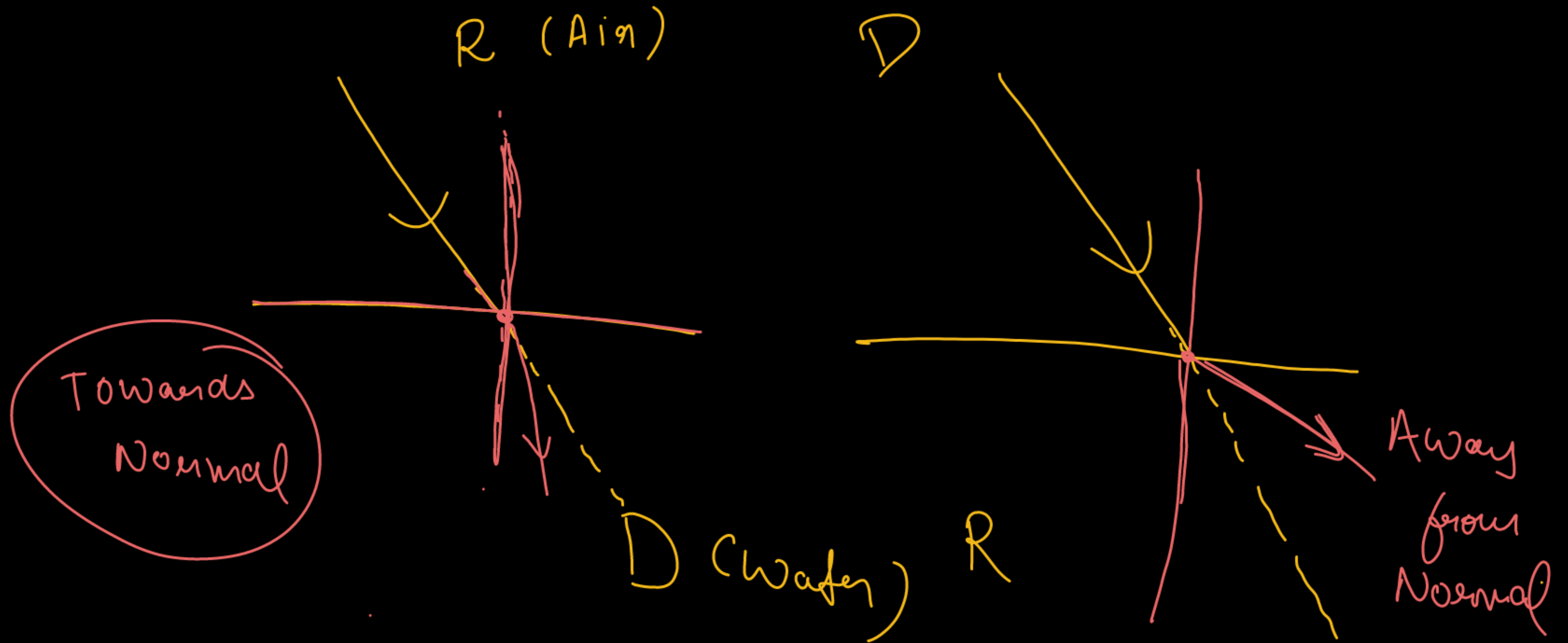
(density kam h)



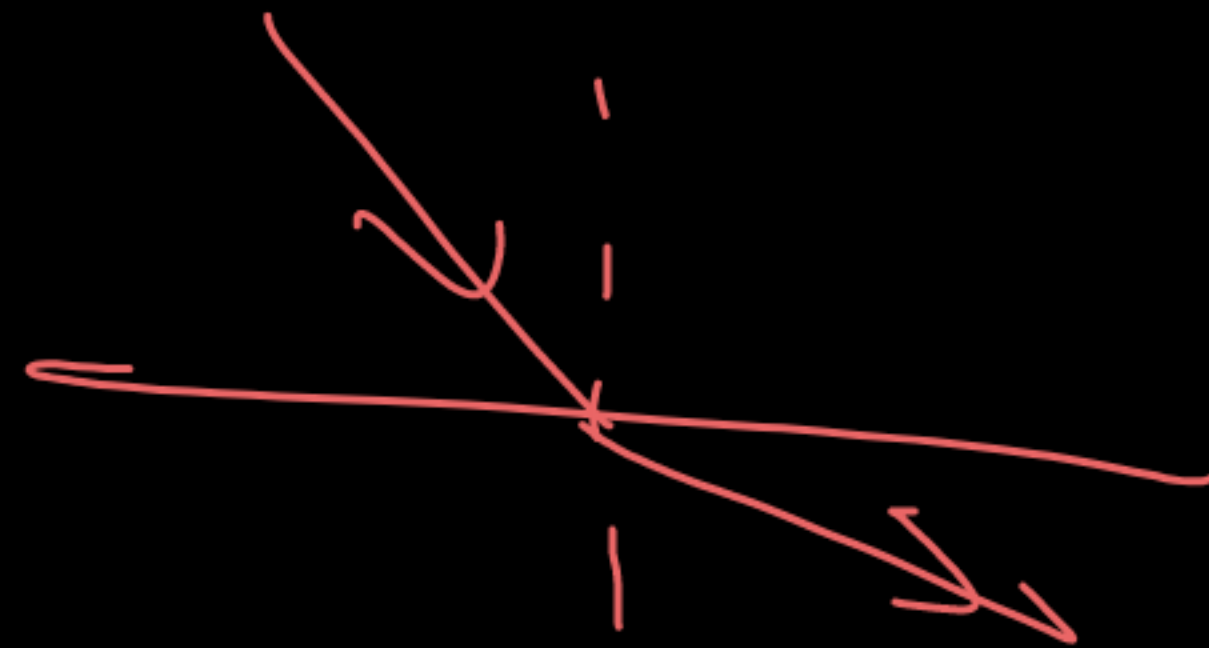
denser



(zyda h)

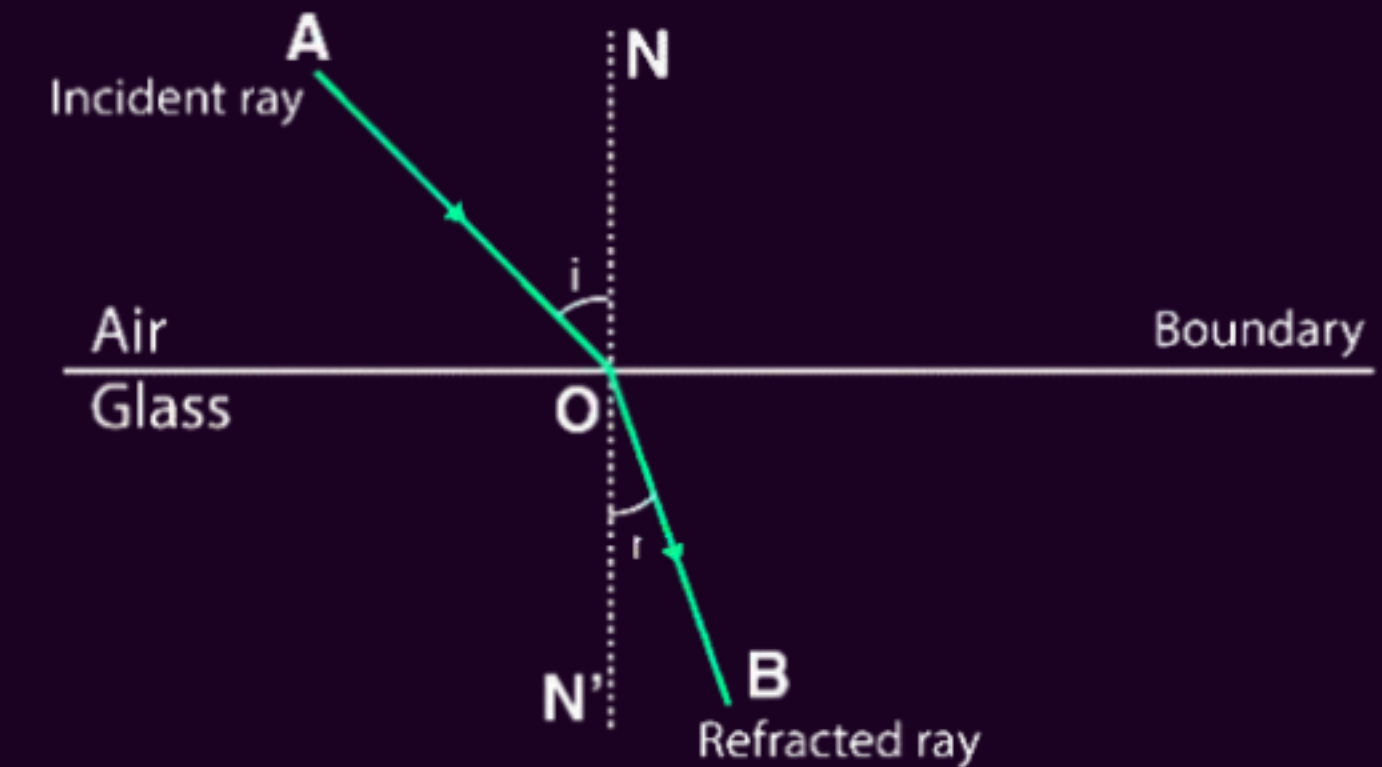


$\textcircled{R} \textcircled{D} \rightarrow \text{Normal}$
 $\textcircled{D} \textcircled{R} \rightarrow \text{nahi h}$



Laws of Refraction

1. The incident ray, the refracted ray, and the normal to the surface all lie in the same plane.
2. The ratio of the sine of the angle of incidence to the sine of the angle of refraction remains constant. It is called as Snell's Law of refraction.

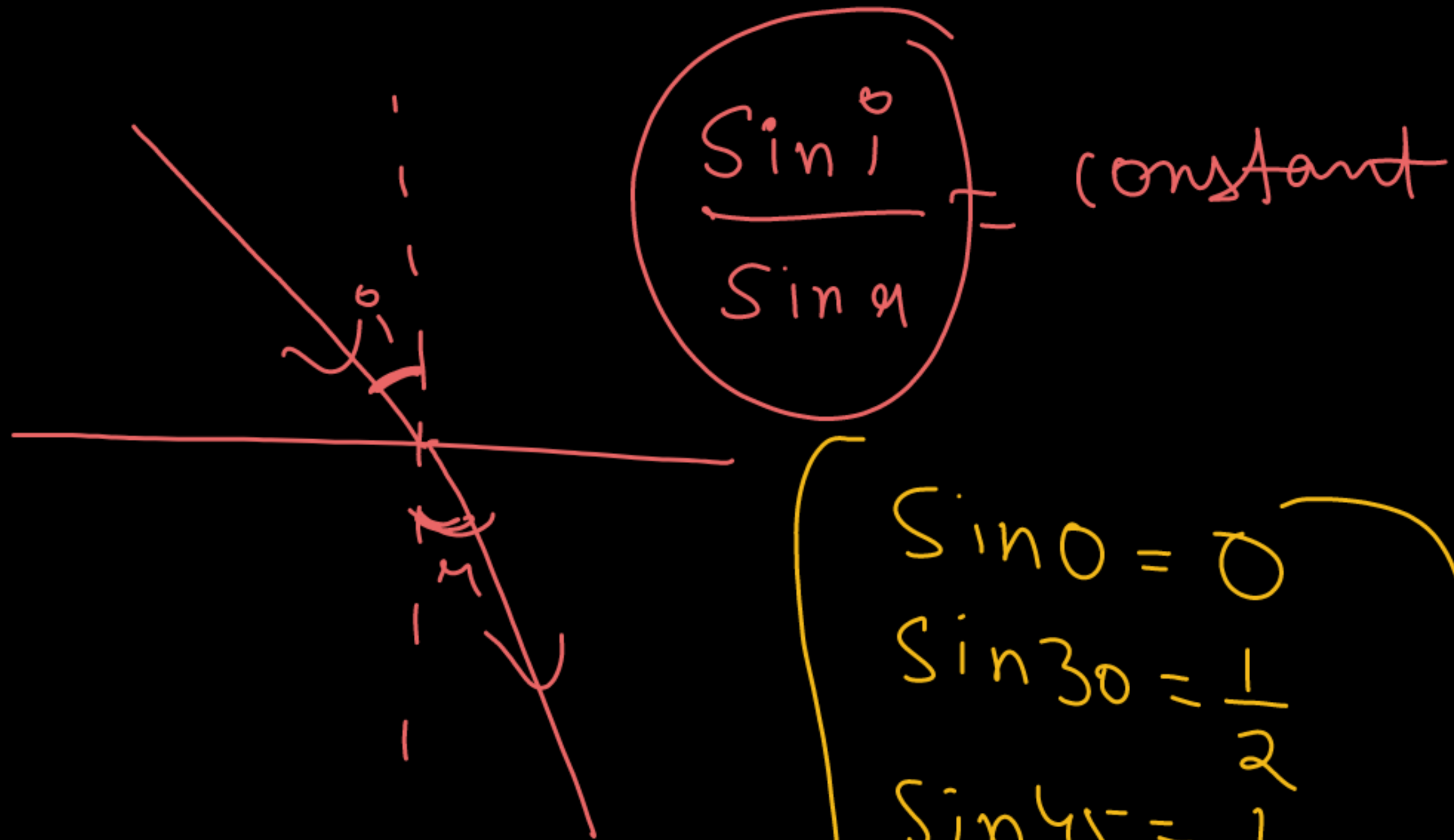


$$\frac{\sin i}{\sin r} = \text{constant}$$

Snell's Law.

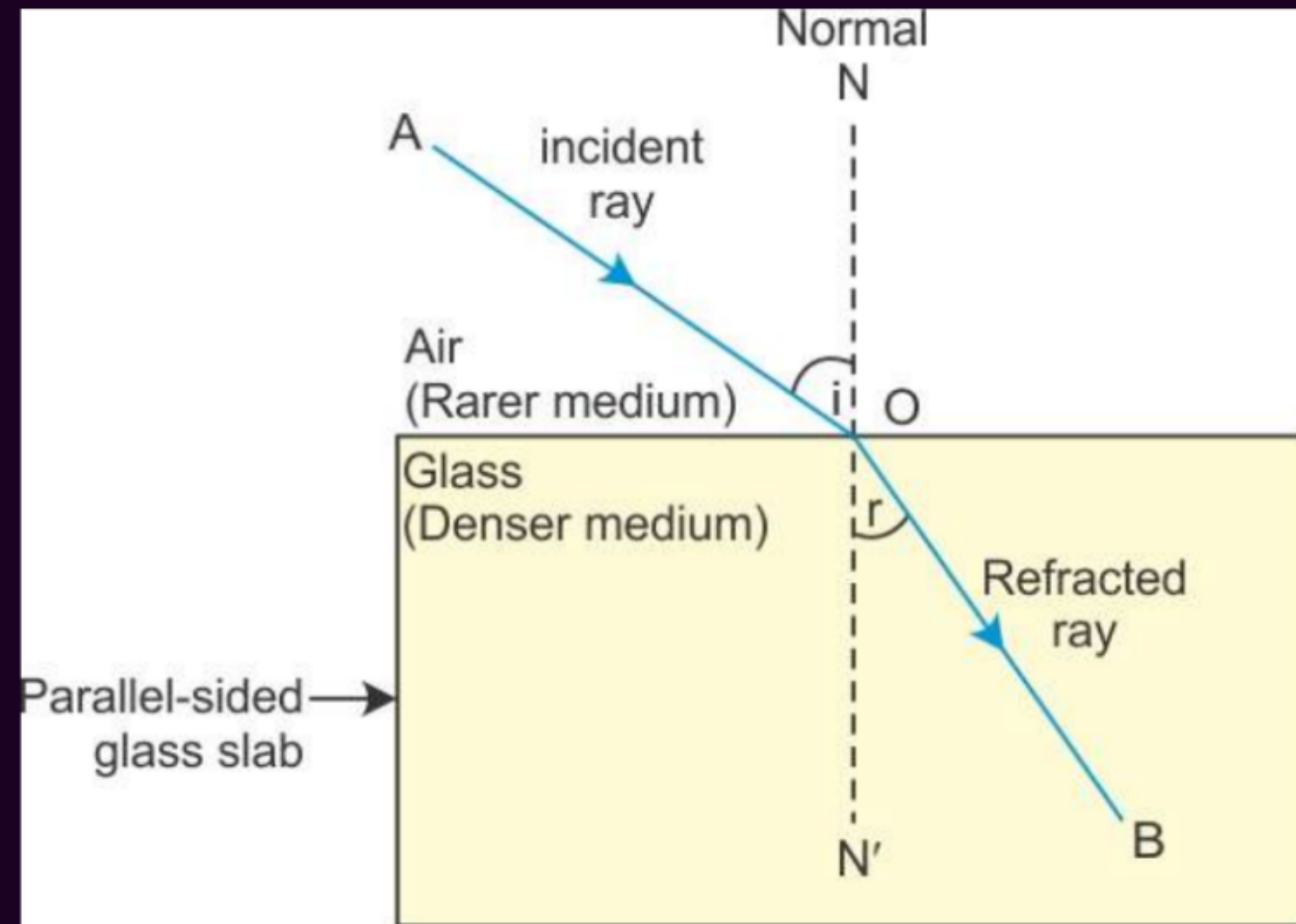


$$\frac{\sin i}{\sin r} = \eta$$

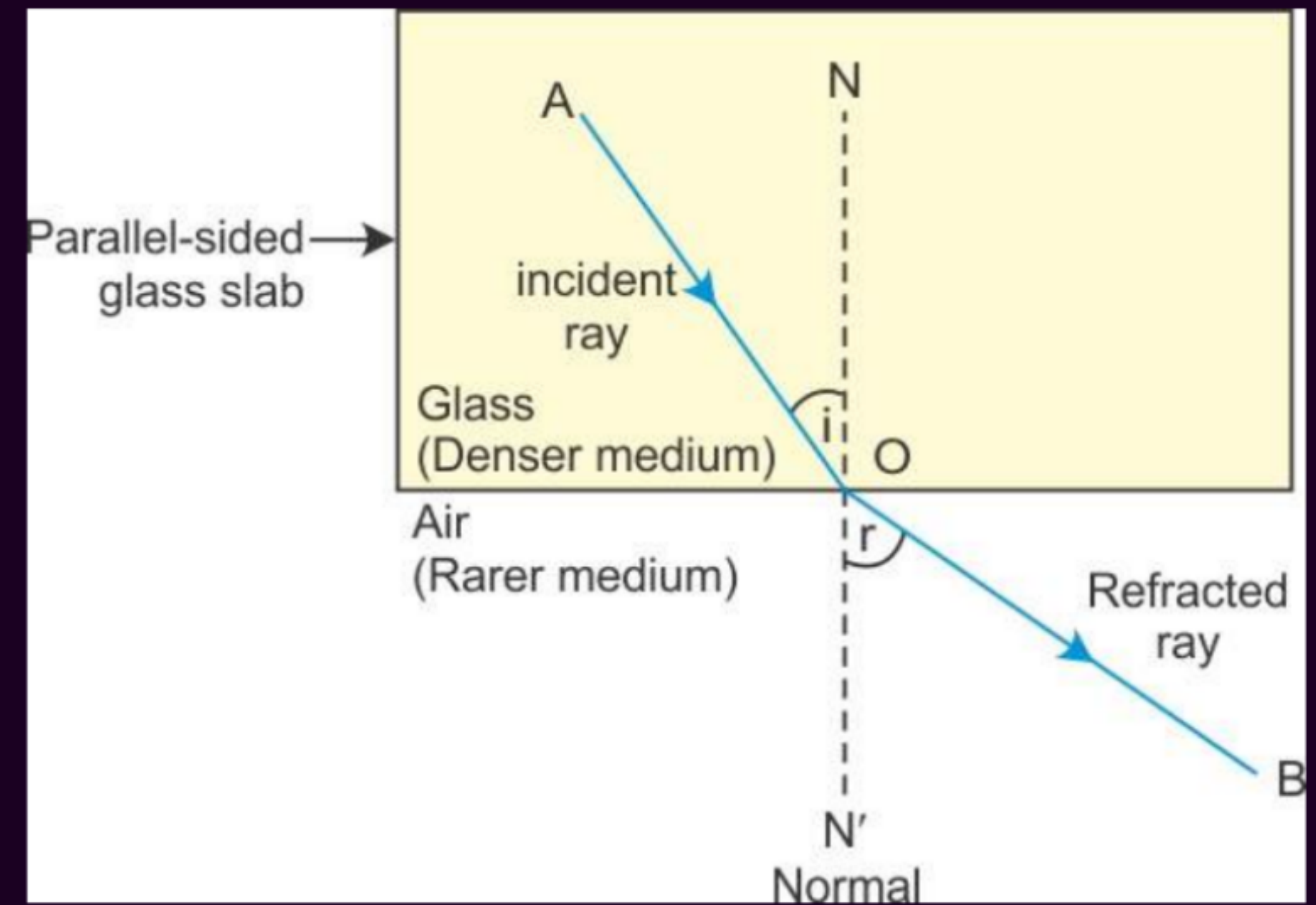


$$\begin{aligned} \sin 0 &= 0 \\ \sin 30 &= \frac{1}{2} \\ \sin 45 &= \frac{1}{\sqrt{2}} \\ \sin 60 &= \frac{\sqrt{3}}{2} \\ \sin 90 &= 1 \end{aligned}$$

Rarer to Denser (Bends towards Normal)



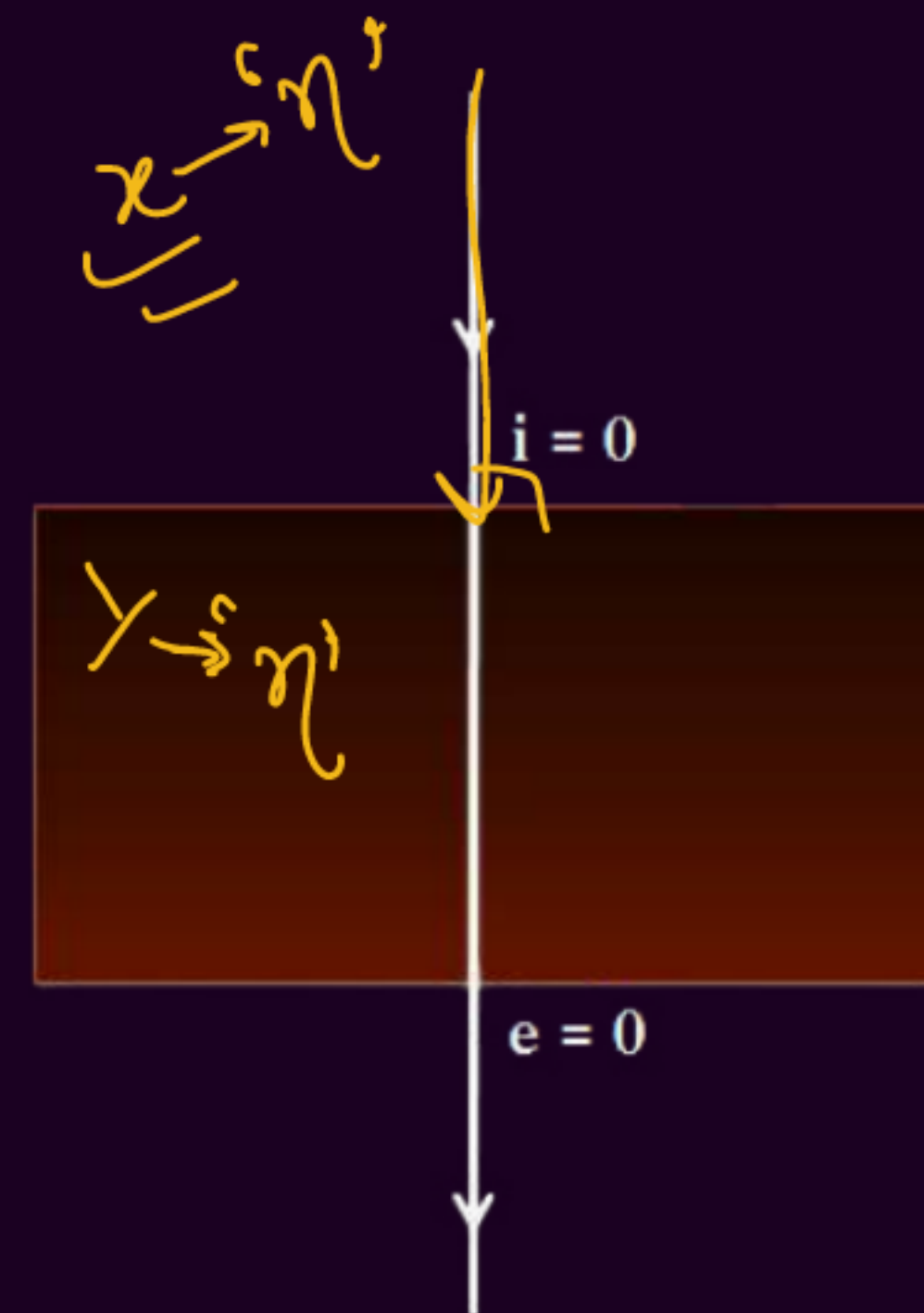
Denser to rarer (Bends away from normal)



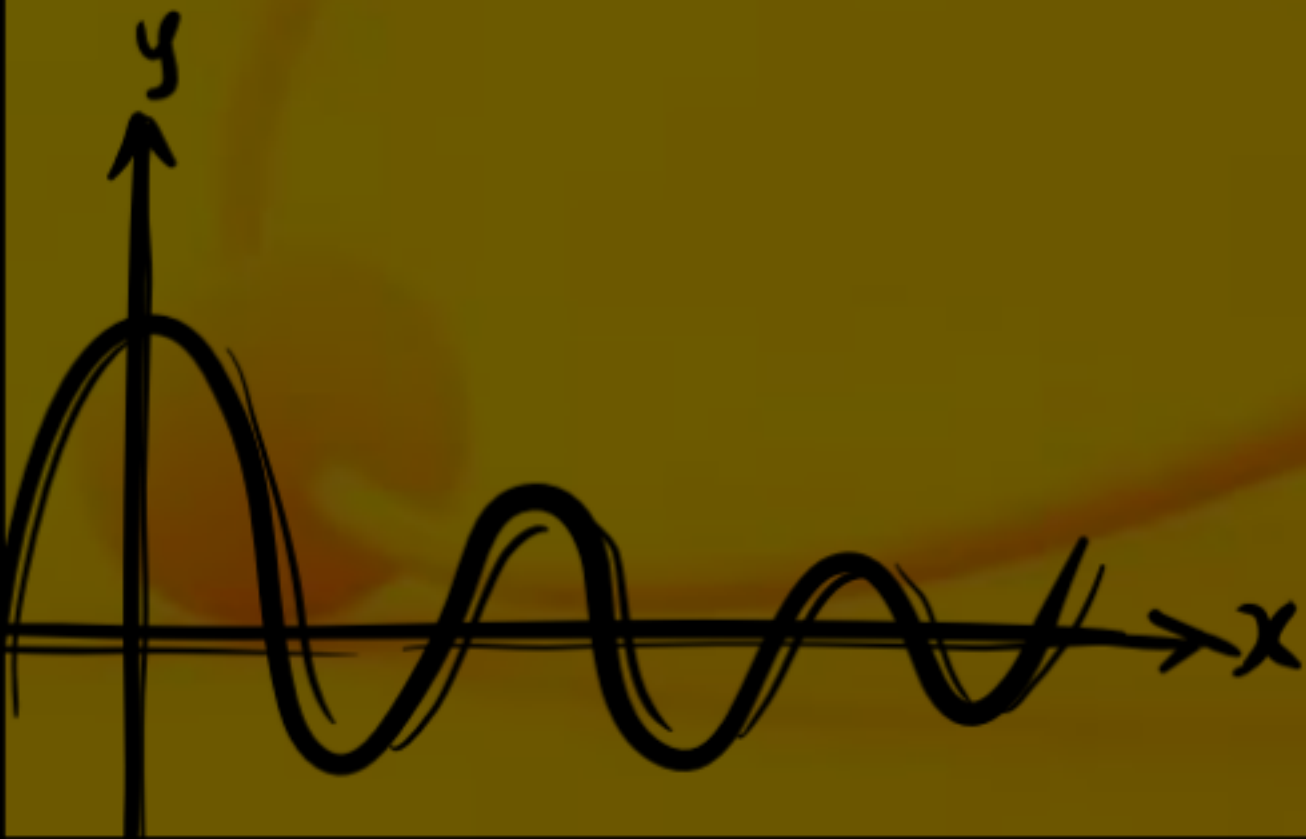
Cases: When bending of light does not occur:

Case I: When light is incident normally

Case II: When light passes through media having the same optical density



Refractive Index



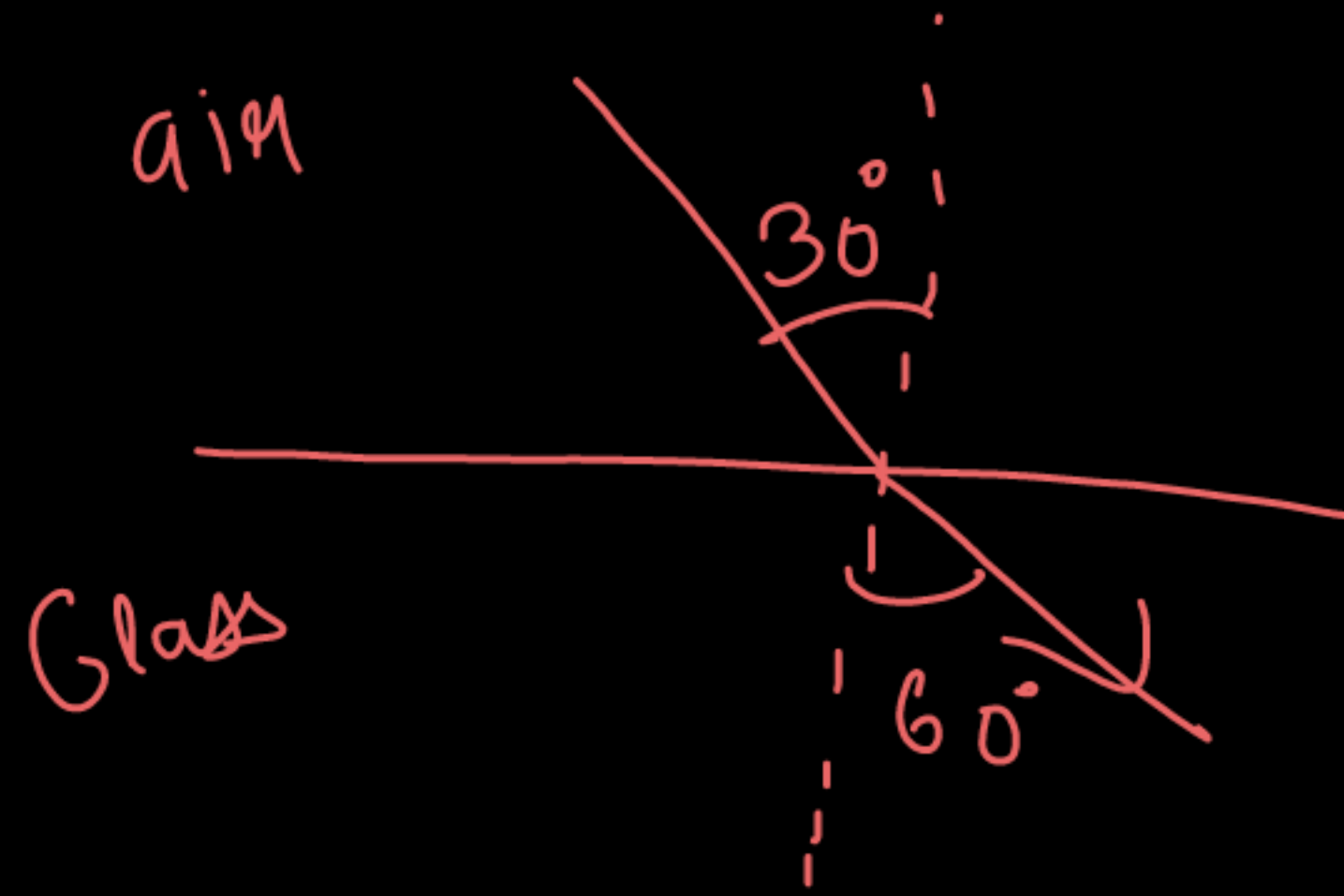
Snell's Law

The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for the given pair of media.

- *Snell's Law describes how light rays bend when they move between different materials.*
- *Equation: $n_1 \sin \theta_1 = n_2 \sin \theta_2$*



$$\frac{\sin i}{\sin r} = \text{constant}$$



$$\frac{\sin i}{\sin r} = n$$

$$\frac{\sin 30}{\sin 60} = n$$

$$n = \frac{\frac{1}{2} \times \frac{2}{\sqrt{3}}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$$

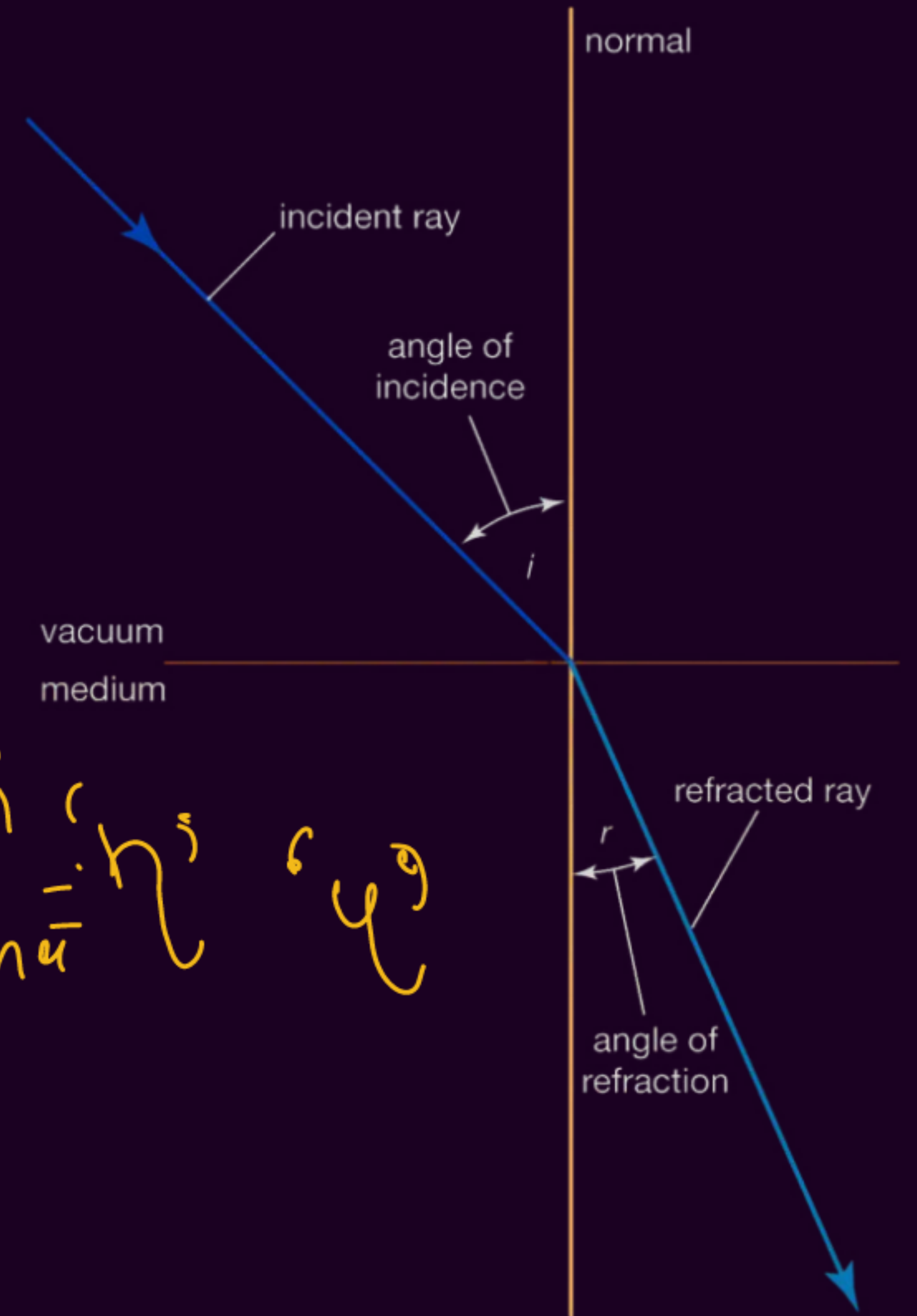
Refractive Index

The refractive index of a medium tells us how much the speed of light decreases when light enters that medium from vacuum/air.

OR

The refractive index (n) of a medium is the ratio of the speed of light in vacuum to the speed of light in that medium.

$$\frac{\sin i}{\sin r} = n$$



$$^c \eta = 1^{\circ} \leq 91.41$$

$$\textcircled{1} [\eta \propto \text{optical density}]$$

$$\textcircled{2} [\eta \propto \text{Bending}]$$

$$\textcircled{3} \boxed{\eta \propto \frac{1}{v}}$$

Formula for η → Unitless

① $\eta = \frac{C}{V}$

→ speed in air ($3 \times 10^8 \text{ m/s}$)

→ speed in that medium

Q → $V = 2 \times 10^8 \text{ m/s}$

$\eta_{\text{water}} = ?$ $\frac{C}{V} = \frac{3 \times 10^8}{2 \times 10^8} = \left(\frac{3}{2}\right)$

Absolute Refractive Index of some Material Media

Material medium	Refractive index	Material medium	Refractive index
Air	1.0003	Canada Balsam	1.53
Ice	1.31	Rock salt	1.54
Water	1.33	Carbon disulphide	1.63
Alcohol	1.36	Dense flint glass	1.65
Kerosene	1.44	Ruby	1.71
Fused quartz	1.46	Sapphire	1.77
Turpentine oil	1.47	Diamond	2.42
Benzene	1.50		
Crown glass	1.52		

Refraction through Glass Slab

When a ray of light passes through a glass slab (a transparent rectangular block), it bends twice:

1. At the first surface (air to glass):

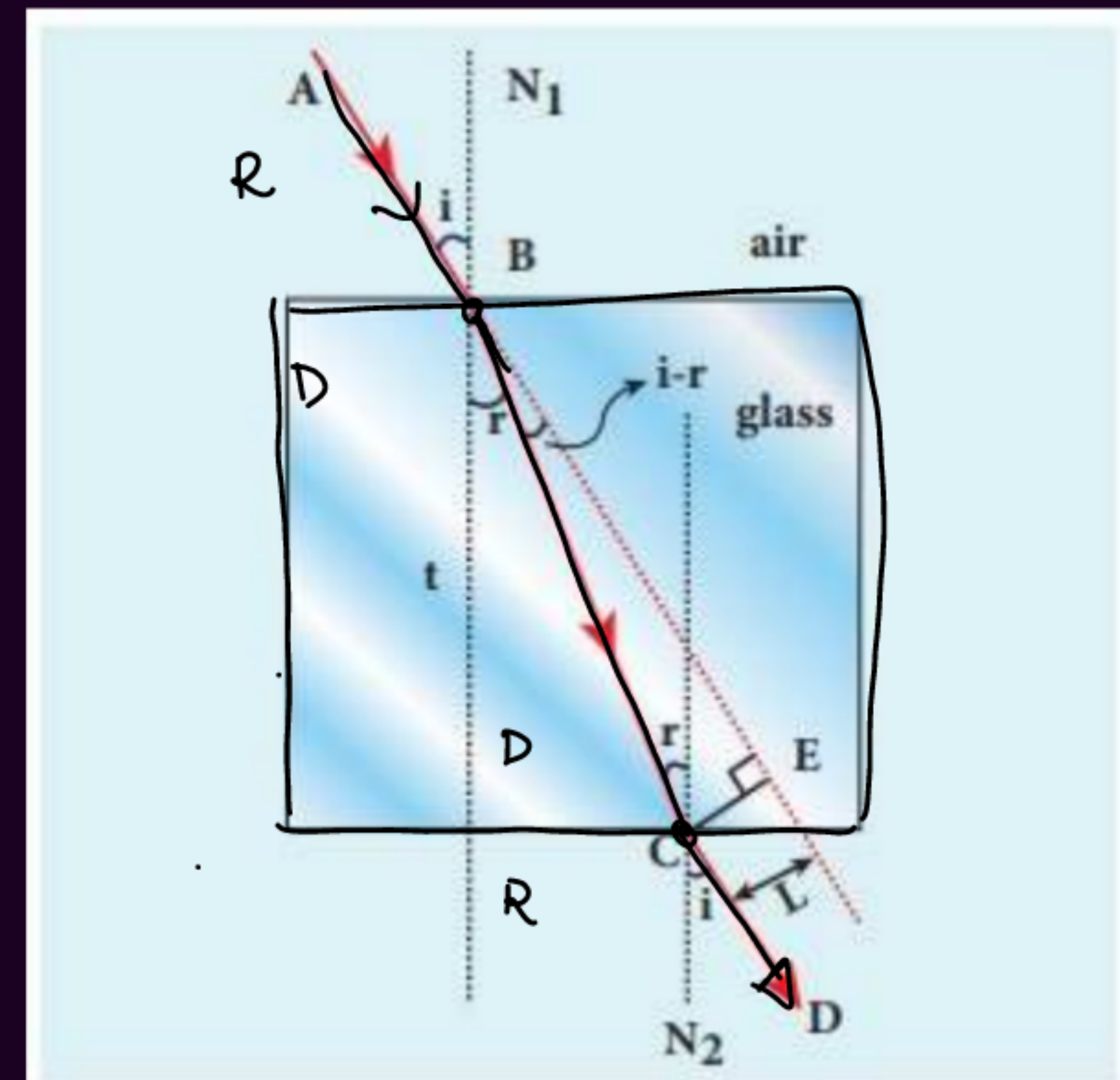
- Light slows down and bends towards the normal.

2. Inside the slab:

- Light travels straight but at a slower speed.

3. At the second surface (glass to air):

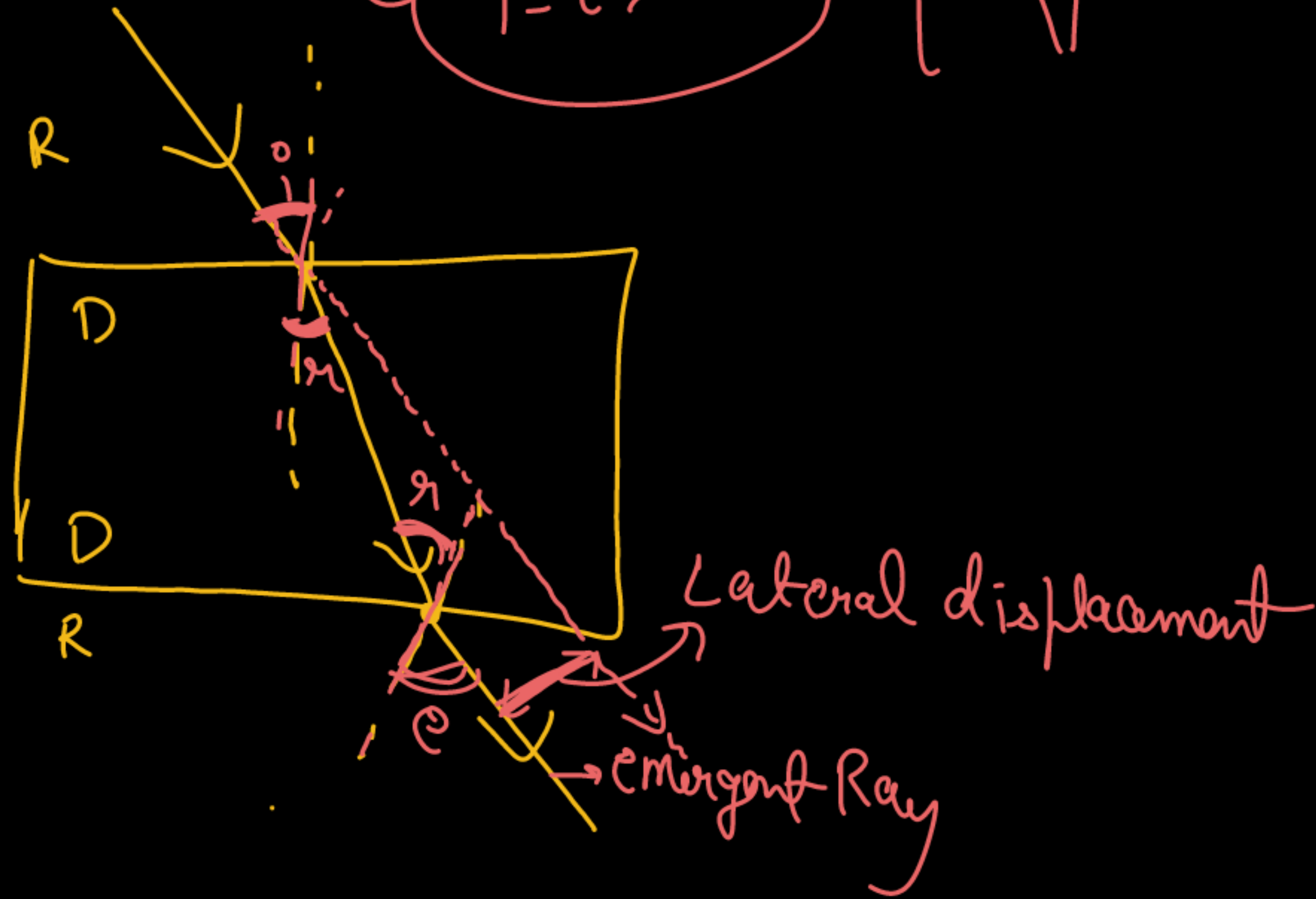
- Light speeds up and bends away from the normal.



glass slab

④ $i = e > r$

AN



The incident ray and the emergent ray (the one coming out) are parallel to each other but shifted sideways.

This sideways shift is called **lateral displacement**.

